

MILAD SOLEYMANI

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[Personal Website](#) – [GitHub](#) – [LinkedIn](#) – [Google Scholar](#)

EDUCATION

- **M.Sc. Computer Science – Artificial Intelligence and Robotics** Sep 2024 – Present
Islamic Azad University, Tehran, Iran
◇ **GPA: 19.27/20 ($\approx 4.0/4.0$)**
- **B.Sc. Electrical Engineering** Sep 2018 – Sep 2023
K. N. Toosi University of Technology, Tehran, Iran
◇ **GPA: 15.78/20 ($\approx 3.16/4.0$)**

RESEARCH EXPERIENCE

- **Language of Reasoning in Large Language Models: Compliance, Code-Switching, and RL Training Dynamics** May 2026 – Present
Supervisor: Dr. Nava Eslami
◇ Demonstrated that architecture, not prompting, sets whether an 8B model reasons in Persian: on 210 paired MathMist problems a reasoning-channel model complied at 4% against 80–85% for inline-CoT models, and disabling its thinking channel flipped the same weights to 84% (paired McNemar)
◇ Showed the chain-of-thought instruction to be near-inert at inference (a 2.9-point language gap) yet decisive under RL, widening to 65–99 points over 300 GRPO steps of a language-blind reward, and that English accuracy falls 91% $\rightarrow$ 33% as Persian-CoT compliance rises, so a model scoring as robust may never have reasoned in Persian
- **Language-Aligned and Parameter-Efficient Representation Learning for Brain Signals** Oct 2025 – Present
Supervisor: Dr. Nava Eslami
◇ Trained a shared EEG/text embedding by aligning an ATCNet encoder with a frozen sentence-transformer (CLIP-style contrastive). The model decodes motor tasks from language descriptions rather than bare labels
◇ Adapted that embedding to new subjects with parameter-efficient fine-tuning techniques. Reinforcement-learning fine-tuning is underway, with the purpose of reliable cross-subject decoding from minimal calibration data
- **Parameter-Efficient Fine-Tuning (PEFT) for Low-Resource Adaptation of Large Vision and Language Models** Jan 2025 – May 2026
Supervisor: Dr. Maliheh Sabeti, Dr. Nava Eslami
◇ Adapted large pretrained vision transformers (Swin, ViT) to specialist medical tasks with parameter-efficient adapters and LoRA. Trained parallel bottleneck adapters on a frozen Swin backbone at 1–8% of model parameters, at accuracy statistically indistinguishable from full fine-tuning, for clinical use on resource-constrained hardware. The resulting efficient Swin architecture for cutaneous melanoma detection is under review

- **Voice Recognition for Chronic Heart Failure (CHF) Phenotyping** Aug 2021 – May 2022
Supervisor: Dr. Andishe Partovi
 - ◇ Developed and evaluated models that detect chronic heart failure from voice recordings. Clinicians defined the phenotypes and validated the outputs
- **Classification and Representation Learning for EEG** Mar 2020 – Jan 2025
Supervisor: Dr. Farhad Goodarzy, Dr. Andishe Partovi
 - ◇ Benchmarked classification and feature-extraction methods, and implemented the time-frequency signal processing behind them
 - ◇ Contributed, as Data Scientist at KeyLead Health, to two EEG projects that resulted in peer-reviewed publications:
 - Partovi A., Burkitt A. N., Grayden D. B. **A Self-Supervised Task-Agnostic Embedding for EEG Signals**, IEEE/EMBS Conference on Neural Engineering (NER), 2023. [IEEE Xplore](#)
 - Partovi A., Grayden D. B., Burkitt A. N. **POC-CSP: A Novel Parameterised and Orthogonally-Constrained Neural Network Layer for Learning Common Spatial Patterns (CSP) in EEG Signals**, Journal of Neural Engineering, 2025. [IOPscience](#)

PUBLICATIONS

Published

- Partovi A., Hosseini S. M., **Soleymani M.**, Liaghat K., Ziaee S., Habibi Panah Fard E., Sadeghpour Vajdi S., Goodarzy F. **A Deep Learning Algorithm for Classifying Grasp Motions using Multi-session EEG Recordings** 2021
9th International Winter Conference on Brain-Computer Interface (BCI)
DOI: [10.1109/BCI51272.2021.9385295](#)

Under review

- Esmkhani M. R., **Soleymani M.**, Ziaee S., Eslami N., Boostani R., Sabeti M. **An Efficient Swin Transformer Architecture for Cutaneous Melanoma Detection** 2026
Biomedical Signal Processing and Control (Elsevier)
Available upon request

In preparation

- **Soleymani M.**, Hoseyni K., Ziaee S., Eslami N. **The Chain-of-Thought Instruction as a Lever on Language Drift: Persian–English GRPO Dynamics under a Language-Blind Reward** 2026
Results complete, manuscript drafted, available upon request
- **Soleymani M.**, Hoseyni K., Ziaee S., Eslami N. **Reasoning in Persian Is Optional: A Compliance Axis for Target-Language Chain-of-Thought in 8B Models** 2026
Results complete, manuscript drafted, available upon request

INDUSTRY EXPERIENCE

- **AI Engineer** **Oct 2025 – Present**
QubitOrbit, UAE / Tehran, Iran
 - ◇ Built typed LLM agents with tool-grounded retrieval and clarification loops that parse domain shorthand into structured records
 - ◇ Led an event-driven market-intelligence pipeline (filter, embed, deduplicate, online-cluster in a vector DB) with a parallel multi-agent analyst
 - ◇ Designed the evaluation harness behind these systems and used it to select prompts and models by measured accuracy and cost
- **AI Engineer** **Sep 2024 – Sep 2025**
MyFast Medical, Australia / Tehran, Iran
 - ◇ Fine-tuned large language models with reinforcement learning for empathic, patient-aligned generation in healthcare
 - ◇ Designed and deployed retrieval-augmented generation pipelines for medical domain tasks
- **Data Scientist (Computer Vision)** **Nov 2023 – Feb 2024**
ATHLX, Canada / Tehran, Iran
 - ◇ Modeled human body movement from video and built classifiers for motion-pattern interpretation
- **Data Scientist (Computer Vision)** **Aug 2022 – Nov 2023**
RUTILEA, Japan / Tehran, Iran
 - ◇ Built a generative pipeline (diffusion, VAE, ControlNet) for high-resolution industrial image generation
- **Data Scientist** **Aug 2021 – Jul 2022**
KeyLead Health, Australia / Tehran, Iran
 - ◇ Built EEG decoding and voice-based phenotyping models (see Research: EEG decoding, chronic heart failure phenotyping)

TEACHING EXPERIENCE

- **Natural Language Processing** **Sep 2025 – Present**
Islamic Azad University, Tehran, Iran — with Dr. Nava Eslami

GRADUATE COURSEWORK

• Deep Learning	20/20	• Deep Reinforcement Learning	19/20
• Large Language Models	20/20	• Evolutionary Computing	18.7/20
• Pattern Recognition	20/20	• Numerical Computation	18.5/20
• Digital Signal Processing	20/20	• Engineering Mathematics	18.4/20
• Machine Learning	19/20	• Advanced Programming	18.2/20

All courses graded 4.0/4.0 on the university's 4-point scale.

COURSES AND CERTIFICATES

- **Task-Oriented Course in Linux**, Quera ([link](#)) **Mar 2023**
- **DeepLearning.AI TensorFlow Developer**, Coursera ([link](#)) **Jun 2021**

- **Advanced Python Programming and Object-Oriented Thinking**, Quera ([link](#)) Jun 2021
- **Deep Learning Specialization**, DeepLearning.AI, Coursera ([link](#)) Aug 2020
- **Machine Learning**, Coursera ([link](#)) May 2020

SKILLS

- **Programming:**

- ◊ Python (expert); NumPy/SciPy/pandas (advanced); Git, Linux, Docker, L^AT_EX (working)

- **Frameworks and Libraries:**

- ◊ PyTorch, TensorFlow/Keras, JAX/Flax, scikit-learn, HuggingFace (Transformers, PEFT, Datasets, Accelerate), PyTorch Lightning, vLLM, XGBoost/LightGBM, OpenCV

- **Machine Learning:**

- ◊ Supervised and unsupervised learning; deep architectures (CNNs, transformers, RNNs, GNNs, diffusion models); representation learning (contrastive, self-supervised)
- ◊ Transfer learning and domain adaptation; parameter-efficient fine-tuning; reinforcement learning (PPO, DPO, GRPO, RLHF); Bayesian and probabilistic modeling; optimization; interpretability and explainability

- **Spoken Languages:**

- ◊ Persian (native), English (fluent — TOEFL iBT 93)

REFERENCES

- **Dr. Nava Eslami:** Researcher, AI Research Center, Sharif University of Technology; Lecturer, Islamic Azad University
Email: nava_eslami_157@iau.ac.ir
- **Dr. Farhad Goodarzy:** Senior Researcher, University of Melbourne, Australia; Senior AI Research Scientist, WEHI (Walter and Eliza Hall Institute of Medical Research)
Email: goodarzy@unimelb.edu.au
- **Dr. Maliheh Sabeti:** Associate Professor, Computer Engineering, Islamic Azad University, North Tehran Branch
Email: sabeti@shirazu.ac.ir
- **Dr. Behrooz Nasihatkon:** Assistant Professor, Computer Science Department, K. N. Toosi University of Technology; CEO, Rahbin Sanat Nasir
Email: nasihatkon@kntu.ac.ir
- **Dr. Andishe Partovi:** AI/ML Specialist, Google Cloud; Cofounder, Veris AI
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